Quantum Computing

Part 7 - Industry, Academia & Claims: Industry & Funding Moves

Compiled by the Spaced Research Ledger

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This part covers the money and the organizations rather than the science. It records who is paying for quantum computing, what governments have committed, which machines national laboratories have actually bought, and what companies have promised to deliver and by when.

One pattern shapes everything below. Private venture funding into quantum startups is slowing, while public money and public markets are expanding rapidly. The sums governments committed in 2026 are larger than the sums venture investors did, which is a reversal from the years before.

Company transitions to public markets, through share offerings and mergers with listed shell companies, are covered separately in Part 8. Section 5 below covers private acquisitions only.

Section 1 covers private funding, Section 2 government programmes, Section 3 national laboratory procurement, Section 4 vendor roadmaps, and Section 5 mergers and acquisitions.

1. Private Funding and Investment

Venture funding into quantum startups is running well below last year's pace, while the amounts raised on public markets by already-listed companies dwarf it. The private round sizes below are small next to the government commitments in Section 2.

Recent Developments

Photon Queue raised $4 million in a seed round led by Playground Global in July 2026, for quantum memory devices [1]. Qolab secured $54.2 million in Series B funding, including convertibles, led by UC Investments in July 2026, to advance superconducting hardware [2]. eleQtron raised a €57 million Series A round in 2026, and stated an order backlog of €54 million [3]. A backlog nearly matching the raise is unusual for a company at that stage.

Current State

The overall trend is a slowdown. Quantum computing startups raised $1.2 billion in seed through growth-stage funding so far in 2026, on track to finish below 2025's record $4.1 billion [5].

Public markets tell the opposite story. Rigetti, D-Wave and Quantum Computing Inc. together raised about $2 billion through public-market financings in the second quarter of 2026 alone [4]. That single quarter from three already-listed companies exceeded the entire private venture total for the year to date.

The largest private rounds came earlier. PsiQuantum raised $1 billion in a Series E led by BlackRock at a $7 billion valuation, the largest single private round in the field's history [7]. Quantinuum raised $600 million to fund a complete platform spanning trapped-ion hardware, software, applications and a fault-tolerance roadmap [4]. QuEra Computing raised over $230 million during 2025, including rounds from Google's quantum division, SoftBank and NVIDIA's venture arm [6].

One investor's behaviour is worth noting on its own. NVIDIA's venture arm invested in Quantinuum, PsiQuantum and QuEra within a single week, spanning all three major hardware approaches [7]. That is a deliberate refusal to pick a winner among the modalities in Parts 1 through 4.

2. Government Programs and Public Spending

Government commitments in 2026 exceeded private investment by a wide margin. The United States, France and Canada each moved in the same period, and the American package is unusual in taking equity rather than only awarding grants.

Recent Developments

The Canadian government committed CAD $195 million, roughly USD $140.2 million, to Xanadu's Project OPTIMISM [8]. The money funds a 158,000-square-foot manufacturing facility in Toronto, announced in August 2026.

The U.S. National Science Foundation awarded $290 million across eight Quantum Leap Challenge Institutes in August 2026 [10]. The awards include renewals, such as $37.5 million for the University of Chicago's institute, alongside new awards focused on sensing, hardware fabrication and fault tolerance. Princeton University will lead one of the eight, targeting hardware fabrication techniques among other goals [9].

Current State

The largest single American commitment came in May 2026. The Department of Commerce announced nine letters of intent providing approximately $2.013 billion in federal incentives under the CHIPS and Science Act to support the domestic quantum industry [13].

The expected distribution concentrates heavily. IBM is expected to receive about $1 billion and GlobalFoundries about $375 million, while D-Wave, Rigetti, Quantinuum and Infleqtion could each receive roughly $100 million [14]. Reports indicate the government would take equity stakes rather than simply awarding grants.

Procurement policy moved alongside the money. The White House issued an executive order in June 2026 to expand federal demand and accelerate commercialization through new procurement models and public-private partnerships [12]. The Defense Innovation Unit has committed up to $200 million to move quantum sensing into military applications [12].

France moved on its own account. President Emmanuel Macron announced an additional €1 billion investment in May 2026 to support domestic quantum capabilities, as part of a technological sovereignty effort [13].

Two American programmes remain unsettled. The House Committee on Science, Space, and Technology approved the National Quantum Initiative Reauthorization Act in April 2026, which would reauthorize federal quantum research programmes and strengthen interagency coordination [13]. The National Science Foundation's National Quantum Virtual Laboratory is still in its design stage [11]. It expects to select the first teams for implementation later in 2026, subject to congressional appropriations.

3. National Laboratory Procurements and Partnerships

National laboratories have shifted from research collaborations to buying machines outright. That change matters more than the sums involved, because a procurement implies an expectation of use rather than of study.

Recent Developments

PsiQuantum signed a $125 million agreement with DARPA under the Quantum Benchmarking Initiative in July 2026, its largest U.S. government award to date [16].

IBM committed up to $50 million in quantum compute access for the U.S. Genesis Mission over five years, announced the same month [17].

HPE and Rigetti announced a partnership to deploy a hybrid quantum-classical system, using a 9-qubit Novera processor, at the Pittsburgh Supercomputing Center testbed [15]. It is funded by a $5 million NSF grant with operations targeted for 2027.

Current State

Oak Ridge National Laboratory deployed a 20-qubit IQM Radiance system named Pathfinder, its first commercially procured quantum computer, installed in its Quantum Computer Deployment Lab for hybrid work [19]. The phrase to notice is first commercially procured. The same machine appears in Part 3 from the vendor's side.

Partnerships continue alongside purchases. IonQ and Sandia National Laboratories signed a memorandum of understanding to explore co-design across computing, networking and national security applications [18]. It expands IonQ's engagement through the Quantum Demonstration Facility in New Mexico. Sandia and Quantinuum have worked together for four years under a research and development agreement renewed in May 2026, and Sandia holds similar agreements with several other companies [21].

Control electronics have their own procurement path. Qblox announced a deployment contract with the Department of Energy and Fermilab, alongside an infrastructure partnership with HPE [20].

Two longer-running structures sit underneath all of this. The Department of Energy's Quantum Systems Accelerator is led by Berkeley Lab with Sandia [23]. It is one of five national quantum information science research centres, established in 2020 and renewed in 2025. Federal agencies including the Department of Energy and the Defense Innovation Unit drive programmes to advance quantum computing for applied science and defence [12].

Canada organizes differently. Its leading quantum research institutes entered a second five-year term of the Quantum Co-laboratory [22]. The network now includes the University of Calgary alongside Waterloo, Sherbrooke and the University of British Columbia.

4. Vendor Roadmaps

Everything in this section is a company statement about the future rather than a result. The dates cluster around 2028 to 2031, and the qubit counts promised are three to four orders of magnitude above anything demonstrated in Parts 1 through 5.

Recent Developments

Diraq published a roadmap paper in August 2026 targeting more than 2 million physical qubits on a single chip by 2031, at a system cost under $1 per qubit [24]. It opened a Santa Monica technology hub the day before, with a team of about 20 and a first product targeted for 2029.

QuEra unveiled plans for a fault-tolerant system with more than 1,000 logical qubits at an error rate of 10^{-9}, targeted for 2028 to 2029 [25]. It would follow the company's Libra system for Amazon Braket in 2028.

Quantinuum announced a multi-year partnership with Oracle to deploy its Helios system as an Oracle Cloud Infrastructure service for hybrid quantum and AI workloads [26].

Current State

The two largest commitments are both for 2030. Quantinuum's published roadmap targets a fully fault-tolerant universal computer, the Apollo system, by 2030, with millions of gates on hundreds of logical qubits [28]. IonQ's roadmap targets 2 million physical qubits and 80,000 logical qubits by 2030, which the company describes as the highest figures any commercial quantum company has committed to [29].

Both are company-stated targets rather than achieved milestones, and should be read against the demonstrated logical-qubit counts in Part 6, which are currently in the tens.

Two other tracks are worth recording. Microsoft's Azure Quantum service offers hardware from Quantinuum, IonQ and Rigetti, while Microsoft's own hardware effort is a separate line developing topological qubits based on Majorana zero modes [27]. Intel's Tunnel Falls roadmap targets thousands of qubits within several years, through a modular architecture combining multiple quantum cores [27]. Its current qubit count remains early relative to competing approaches.

5. Mergers and Acquisitions

The buying pattern here is vertical. Quantum companies are acquiring their own suppliers, particularly in semiconductor packaging and photonic components, rather than acquiring each other.

Current State

Quantum Computing Inc. made three acquisitions in five months. It completed its purchase of Luminar Semiconductor, a photonic components manufacturer, in an all-cash transaction valued at $110 million in February 2026 [32]. It completed its acquisition of NuCrypt, a quantum communications company, for $5 million in March 2026 [31].

In June 2026 it completed the purchase of NHanced Semiconductors, a U.S. semiconductor packaging foundry [30]. The price was $73.1 million in cash and stock plus up to $72.0 million in earnouts, and it launched the company's Fab 2 initiative.

Buying a packaging foundry is a supply-chain move rather than a technology one.

The largest transaction was different in kind. D-Wave entered an agreement to acquire Quantum Circuits Inc., a developer of error-corrected superconducting gate-model systems, for $550 million [33]. The price splits into $300 million in D-Wave stock and $250 million in cash. The combined company plans to bring superconducting gate-model systems to market in 2026. D-Wave's own machines use a different approach, so this buys a capability the company did not have.

Three public-market transitions also occurred in this period and are covered in Part 8. Quantinuum completed a traditional initial public offering raising $1.7 billion in gross proceeds [26]. Pasqal announced it would go public through a merger with a listed shell company at a $2 billion pre-money valuation, including $200 million in committed convertible financing [34]. IQM announced a similar transaction in February 2026 [34].

About This Report

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Sources

1. Photon Queue raises $4m in Seed round for quantum memory devices - https://www.datacenterdynamics.com/en/news/photon-queue-raises-4m-in-seed-round-for-quantum-memory-devices/ (as of 2026-07-23)
2. Qolab Secures $54.2M Series B Funding for Quantum Hardware Development - https://www.hpcwire.com/off-the-wire/qolab-secures-54-2m-series-b-funding-for-quantum-hardware-development/ (as of 2026-07-02)
3. eleQtron secures €57 million in Series A funding round - https://thequantuminsider.com/2026/05/05/eleqtron-secures-e57-million-in-series-a-funding-round/ (as of 2026-05-05)
4. Quantum Computing Market Fundraising Deals (2026) - https://newmarketpitch.com/blogs/news/quantum-computing-funding-deals (as of 2026-06-30)

5. Sector Snapshot: Quantum Computing Startup Investment Slows In 2026 While Public Markets Hold Strong - https://news.crunchbase.com/venture/quantum-computing-startup-investment-data-quantinuum-ipo/ (as of 2026-05-18)
6. Most Funded Quantum Computing Startups 2026 - https://entangledfuture.com/guides/most-funded-quantum-startups/ (as of 2025-12-31)
7. Top Quantum Computing Investors in 2026 - https://thequantuminsider.com/2026/06/26/top-quantum-computing-investors-in-2026/ (as of 2025-09-01)
8. Xanadu secures CAD 195 million and establishes landmark advanced photonics facility - https://www.globenewswire.com/news-release/2026/08/28/3352847/0/en/xanadu-secures-cad-195-million-and-establishes-landmark-advanced-photonics-facility.html (as of 2026-08-28)
9. Princeton to lead major federally funded institute to unlock quantum computing challenges - https://engineering.princeton.edu/news/2026/08/25/princeton-lead-major-federally-funded-institute-unlock-quantum-computing-challenges (as of 2026-08-25)
10. NSF Awards $290M Across Eight Quantum Leap Challenge Institutes - https://quantumcomputingreport.com/nsf-awards-290m-across-eight-quantum-leap-challenge-institutes-renews-uchicago-qubbe-with-37-5m/ (as of 2026-08-25)
11. NSF selects five additional teams in National Quantum Virtual Laboratory design competition - https://www.nsf.gov/news/nsf-selects-five-additional-teams-national-quantum-virtual (as of 2026-06-24)
12. White House Accelerates Quantum Technology Procurement - https://app.govly.com/public/signals/134686 (as of 2026-06-22)
13. QED-C Law and Policy Briefing, June 2026 - https://quantumconsortium.org/publication/law-and-policy-briefing-june-2026/ (as of 2026-05-22)
14. Reports: US to Award $2 Billion to Quantum Companies, Take Equity Stakes - https://thequantuminsider.com/2026/05/21/reports-us-to-award-2-billion-to-quantum-companies-take-equity-stakes/ (as of 2026-05-21)
15. HPE, Rigetti partner for hybrid quantum-classical computer deployment at Pittsburgh Supercomputing Center testbed - https://www.datacenterdynamics.com/en/news/hpe-rigetti-partner-for-hybrid-quantum-classical-computer-deployment-at-pittsburgh-supercomputing-center-testbed/ (as of 2026-07-28)
16. Quantum computing startup PsiQuantum signs $125 million DARPA deal - https://www.reuters.com/technology/quantum-computing-startup-psiquantum-signs-125-million-darpa-deal-2026-07-22/ (as of 2026-07-22)
17. IBM commits $50M in quantum access for US Genesis Mission - https://research.ibm.com/blog/ibm-us-genesis-mission-quantum-ai (as of 2026-07-22)
18. IonQ and Sandia National Laboratories Partner on Quantum Technology Research - https://thequantuminsider.com/2026/08/05/ionq-sandia-quantum-technology-research-partnership/ (as of 2026-08-05)
19. ORNL Deploys New IQM Quantum Computer - https://www.olcf.ornl.gov/2026/07/08/ornl-deploys-new-iqm-quantum-computer/ (as of 2026-07-08)

20. Qblox Finalizes Crucial U.S. National Laboratory Licensing Agreement - https://quantumcomputingreport.com/qblox-finalizes-crucial-u-s-national-laboratory-licensing-agreement-and-expands-enterprise-high-performance-computing-footprint/ (as of 2026-06-26)
21. In the Mountain West, a quantum computing collaboration announces major results - https://www.sandia.gov/labnews/2026/07/20/in-the-mountain-west-a-quantum-computing-collaboration-announces-major-results/ (as of 2026-05-01)
22. Quantum Co-laboratory Extends Five-Year National Collaboration - https://thequantuminsider.com/2026/02/21/quantum-co-laboratory-five-year-extension/ (as of 2026-02-21)
23. Partnering with Industry to Accelerate Quantum Computing - https://newscenter.lbl.gov/2026/05/28/partnering-with-industry-to-accelerate-quantum-computing/ (as of 2025-01-01)
24. Diraq opens US technology hub in Santa Monica - https://thequantuminsider.com/2026/08/26/diraq-opens-us-technology-hub-santa-monica/ (as of 2026-08-27)
25. QuEra Unveils Gigaquop-Class Fault-Tolerant Roadmap - https://www.prnewswire.com/news-releases/quera-unveils-gigaquop-class-fault-tolerant-roadmap-and-invites-organizations-to-co-design-quantum-applications-302810334.html (as of 2026-06-25)
26. Quantinuum Reports Second Quarter 2026 Results - https://www.quantinuum.com/press-releases/quantinuum-reports-second-quarter-2026-results (as of 2026-08-11)
27. The Companies Building Quantum Computing Chips in 2026 - https://thequantuminsider.com/2026/06/05/how-many-quantum-chip-companies-are-there/ (as of 2026-06-25)
28. Quantum Roadmaps Reviewed - https://brianlenahan.substack.com/p/quantum-roadmaps-reviewed (as of 2024-09-01)
29. IonQ Roadmap - https://www.ionq.com/roadmap (undated)
30. Quantum Computing Inc. Form 8-K, NHanced Semiconductors - https://www.sec.gov/Archives/edgar/data/0001758009/000121390026070872/ea029546601ex99-1.htm (as of 2026-06-23)
31. Quantum Computing Inc. Form 8-K, NuCrypt - https://www.sec.gov/Archives/edgar/data/1758009/000121390026023896/ea028013801ex99-1.htm (as of 2026-03-05)
32. Quantum Computing Inc. Form 8-K, Luminar Semiconductor - https://www.sec.gov/Archives/edgar/data/1758009/000121390026011452/ea027538301ex99-1_quantum.htm (as of 2026-02-02)
33. D-Wave Announces Agreement to Acquire Quantum Circuits Inc. - https://ir.dwavequantum.com/news/news-details/2026/D-Wave-Announces-Agreement-to-Acquire-Quantum-Circuits-Inc–Establishing-Worlds-Leading-Quantum-Computing-Company/default.aspx (undated)

34. Quantum computing company Pasqal to go public via SPAC merger - https://www.datacenterdynamics.com/en/news/quantum-computing-company-pasqal-to-go-public-via-spac-merger/ (as of 2026-03-05)